VULCAN

AI-Powered Daily Reports for Construction

Turn jobsite chaos into professional documentation — in minutes, not hours.

Pi Hacks 2026 | Healthcare / HealthTech Track | University of Michigan

github.com/montben/VulcanAI

THE PROBLEM

Documentation is construction's blind spot

> *"End-of-day reports are scattered across phone photos, texts, memory, and sometimes handwritten notes on someone's arm."*
> — Vulcan Team Member, 2 years in construction

85%
of contractors spend >25% of their time on tracking & reporting

$31.3B
lost annually to rework caused by poor documentation

52%
of construction rework driven by bad data & miscommunication

Small contractors are hit hardest — no admin staff, no enterprise software, no time after a 12-hour day.

MARKET OPPORTUNITY

A massive, fragmented industry

$1.27T

U.S. Construction Market (2025)

Growing 5.6% YoY

814K+

Construction Businesses with Employees

Highly fragmented market

91 hrs

Per Employee Per Year Spent on Paperwork

Reading, filling, searching

#1

Cause of Disputes is Poor Documentation

Poorly drafted claims

No single company holds >5% market share. Small firms dominate — and they have the fewest tools.

THE SOLUTION

Talk through your day. Get a professional report.

Voice Check-In

AI-guided voice call asks targeted questions about crew, deliveries, safety, and tomorrow's plan. Designed for the drive home.

Photo Upload

Drag-and-drop jobsite photos with captions. Visual documentation preserved alongside the narrative report.

Branded PDF

Polished daily report with completed work, crew activity, delays, safety notes, weather, and attached photos.

No typing. No templates. No desk work. Just talk and upload.

HOW IT WORKS

A multi-step AI pipeline, not a simple chatbot

1

Voice Intake

Deepgram live agent asks targeted questions

2

Transcription

Groq Whisper converts speech to text

3

Photo Context

Jobsite photos & captions attached

4

Report Synthesis

LLaMA 3.3 merges transcript + metadata + weather

5

PDF Generation

Branded report ready to download or send

Weather data automatically enriched using project location — the final output feels like a real construction record.

TECH STACK

Built for speed, reliability, and real-world use

FRONTEND

- React 19 + TypeScript
- Vite build system
- Tailwind CSS + shadcn/ui
- TanStack Query
- Drag & drop uploads
- In-browser PDF viewer

BACKEND

- Python + FastAPI
- PostgreSQL + SQLAlchemy
- ReportLab PDF generation
- Custom branded layout
- Custom font typography
- Weather API integration

AI PIPELINE

- Deepgram voice agent
- Groq Whisper transcription
- Groq LLaMA 3.3 synthesis
- Multi-step orchestration
- Structured data extraction
- Context-enriched output

WHY VULCAN WINS

Designed for the end of a 12-hour day

WITHOUT VULCAN	WITH VULCAN
✗ Photos lost in camera roll	✓ Photos uploaded & captioned on-site
✗ Notes scribbled on paper or texts	✓ AI-guided voice call on the drive home
✗ Reports done from memory hours later	✓ Structured report generated in minutes
✗ No consistent format or paper trail	✓ Branded, professional PDF every day
✗ Disputes become he-said-she-said	✓ Dated records for billing & disputes

CHALLENGES & LEARNINGS

Building trust into every step of the pipeline

Reliability Over Demo Magic

Easy to make an LLM summarize once — hard to produce structured, trustworthy output that holds up as a real record, every time.

Voice UX Is Its Own Problem

Tuned interrupt timing, response speed, and pacing so the call feels natural while still extracting structured data (crew counts, materials, safety).

Multi-Agent Orchestration

Each pipeline stage has its own job and context. The final report is richer than any single-pass prompt could produce alone.

Built for Exhaustion

Every UX decision filtered through: 'Would someone do this after a 12-hour day?' Upload photos on-site, call on the drive home, done.

WHAT'S NEXT

From hackathon to the field

1	**Mobile-First Capture**	Native mobile experience for on-site photo upload and voice check-in
2	**Multilingual Support**	Support crews who work across languages on the same jobsite
3	**Smarter Photo Analysis**	AI-powered photo understanding for automatic progress detection
4	**Report Editing & Sharing**	Edit before finalizing, then share via email or PM tools
5	**Real Contractor Testing**	Partner with small contractors to refine around real workflows

Vision: A lightweight field documentation layer that gives small teams the consistency and protection that larger firms get from enterprise systems.

VULCAN

Talk. Upload. Done.

AI-powered daily reports for the contractors who need them most.

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